

Summary

Adaptive Localization-Aware Goal-Directed Navigation for 2D LiDAR Mobile Robots

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1 Proposed Localization-Adaptive Navigation Mechanism

The proposed system achieves localization-aware goal-directed exploration by continuously estimating both the robot’s current localization uncertainty and the localizability of the surrounding environment while the robot builds the map using SLAM.

First, the robot uses its LiDAR observations and the partially constructed SLAM map to estimate how informative different regions of the environment are for localization. A Fisher-information-based localizability metric is used to identify regions with strong geometric constraints, such as walls, corners, and other distinctive structures, as well as weakly localizable regions such as large open or geometrically ambiguous spaces. This information is converted into a localizability or localization-risk layer that can be used by the navigation planner.

At the same time, the robot continuously monitors its current pose covariance obtained from the SLAM system. The covariance represents the robot’s present level of uncertainty about its estimated position. Instead of treating localization risk as a fixed navigation cost, the covariance is used as feedback to dynamically determine how important the localizability of the environment should be during planning.

When the robot has low localization uncertainty, the localization-risk penalty is kept relatively small. In this condition, the planner can prioritize progress toward the goal and may choose a shorter path even if part of that path passes through a moderately weakly localizable region.

As the robot’s covariance increases, however, the localization-risk weight is increased. The planner then places greater importance on the Fisher-based localizability information and begins to prefer paths containing stronger geometric features. Consequently, the robot may temporarily take a slightly longer route near walls, corners, or other informative structures in order to reduce the probability of further localization degradation.

This creates a closed-loop relationship between localization and navigation. The planner does not simply follow a predefined localization-safe path. Instead, the robot continuously observes its localization state, evaluates the geometry available in the partially constructed map, and adjusts its navigation priorities accordingly.

The overall path cost can be represented as

$$J = J_{\text{goal}} + \lambda(\Sigma)J_{\text{localization}} + J_{\text{navigation}}, \quad (1)$$

where J_{goal} represents the cost associated with making progress toward the destination, $J_{\text{localization}}$ represents the localization risk derived from the Fisher-information-based localizability map, and $J_{\text{navigation}}$ represents conventional navigation costs such as obstacle avoidance and path length. The term Σ denotes the robot's current pose covariance, while $\lambda(\Sigma)$ is an adaptive localization-risk weight that changes according to the robot's current localization uncertainty.

The adaptive behavior can be expressed conceptually as

$$\Sigma \downarrow \Rightarrow \lambda(\Sigma) \downarrow, \quad (2)$$

which causes the planner to place greater emphasis on efficient progress toward the goal. Conversely,

$$\Sigma \uparrow \Rightarrow \lambda(\Sigma) \uparrow, \quad (3)$$

which increases the importance of localization quality and encourages the planner to favor regions containing stronger geometric information.

Therefore, the objective of the proposed method is not to force the robot to remain in highly localizable regions at all times. Instead, the robot is allowed to traverse weakly localizable regions when its current localization state is sufficiently reliable. As localization uncertainty increases, the navigation system gradually becomes more conservative and prioritizes regions that provide stronger localization information.

In this way, the proposed system dynamically balances goal progress, exploration, navigation efficiency, and localization robustness while simultaneously constructing the map and navigating toward the final destination.

1.1 System Architecture

The overall architecture of the proposed system can be summarized as follows:

1. LiDAR and odometry measurements are provided to the SLAM system.
2. The SLAM system generates a partially constructed map and an estimate of the robot's current pose and pose covariance.
3. The LiDAR observations and partial map are used to compute a Fisher-information-based localizability layer.
4. The current pose covariance is used to determine the adaptive localization-risk weight $\lambda(\Sigma)$.
5. The Fisher-based localization-risk layer is weighted according to the current localization uncertainty.
6. The weighted localization-risk cost is combined with conventional navigation costs and goal-directed costs.
7. The global planner computes a path that balances progress toward the destination with localization robustness.

8. As the robot moves, the map, covariance, localizability layer, and navigation costs are continuously updated, forming a closed-loop localization-aware navigation system.

The resulting feedback process can be summarized as

Localization uncertainty increases → Covariance increases
→ Localization-risk weight increases
→ Planner favors informative geometry
→ Localization robustness improves. (4)

Once the localization uncertainty becomes sufficiently low again, the localization-risk weight decreases and the planner can place greater emphasis on direct progress toward the final goal. ““